UNIVERSITEIT·STELLENBOSCH·UNIVERSITY
jou kennisvennoot • your knowledge partner

Sales data analysis for statistical process control

Andre van Zyl

23710608

27/10/2023

Table of Contents

Introduction 4

Methodology 4

Data analysis techniques applied to the database 4

Section 1 5

Data wrangling 5

Section 2 6

Descriptive statistics 6

Descriptive statistics of Age 7

Distribution of age over the departments 7

Descriptive statistics of Delivery time 8

Distribution of delivery times 8

Descriptive statistics of price 9

Distribution of price 9

Section 3 10

Control charts for various departments 10

eBooks 11

Technology 12

Capability measures 13

Capable Departments 14

Process out of and under control 14

Section 4 15

Type I Error in Control Charts 15

Considering the specific scenarios outlined: 15

A. Single Xbar Sample Beyond Outer Control Limits: 15

B. Consecutive Xbar Samples Between -1 and +1 Sigma-Control Limits: 15

Optimising delivery times 16

Normal distribution of unadjusted mean 16

Normal distribution of adjusted mean 16

Visualization of minimum cost 17

Optimal reduction in hours and total minimum cost 17

Section 5 18

Single factor experiment 18

All departments 18

Luxury 18

Least significant difference 19

All departments 19
Paper 19
Section 6 20
Reliability of service 20
Optimal hiring solution for minimum cost 20
P-value 21
Results of optimal Solution 21
Conclusion 21
References 22
Appendix 22
Luxury 22
Paper 22
Art 23
Books 24
Envelopes 25
Accessories 25
Stationary 26

Introduction

Statistical process control (SPC) is a pivotal tool in ensuring the consistent quality and efficiency of production processes. In today's competitive market, the margin for error is minimal, and even slight deviations can lead to significant financial losses or damage to a company's reputation. SPC provides a systematic method for monitoring, controlling, and optimizing processes by analysing performance data. By identifying variations in processes, SPC allows organizations to pinpoint issues before they escalate into larger problems, ensuring products or services meet consistent standards. Moreover, it aids in reducing waste, improving productivity, and enhancing customer satisfaction. Without SPC, businesses risk operating blindly, relying on reactive measures rather than proactive oversight. In essence, SPC is not just a quality assurance tool but a necessity for organizations aiming for operational excellence and a competitive edge.

Methodology

Data analysis techniques applied to the database

The analysis of this project was done by meticulously examining a substantial database consisting of 120,000 records. Our primary approach involved sorting the data, followed by a comprehensive graphical representation.

The purpose of these graphs is to provide a visual insight into various aspects of our operations. Specifically, they aim to:

- Highlight which products outperform others in terms of sales or other relevant metrics.

- Understand the underlying reasons for sustained customer loyalty.

- Identify potential areas of improvement or concern.

By leveraging these insights, we hope to make informed decisions that will drive our business forward, ensuring both product excellence and continued customer satisfaction.

Section 1

Data wrangling

Upon loading the data set it was analysed for missing values and two new data sets were created, one containing all the complete instances and one with all the incomplete instances, after reviewing both new sets it was clear that there were no instances that contained missing values.

Next, we dove into the meaning of the contents within the data set and found the following column headings:

- ID: Represents a specific customer.
- AGE: Age of the customer.
- Department: 1 of 9 departments from which goods are purchased.
- Supervisor: contains the string “ERR” since we do not have authorised access to the contents.
- Price: Price of the goods purchased.
- Year: year in which the transaction took place.
- Month: Month in which the transaction took place.
- Day: Day on which the transaction took place.
- Ordertime: Time of order placement.
- Deliverytime: Time it took to deliver an Order.
- Partner: contains the string “ERR” since we do not have authorised access to the contents.

Section 2

Descriptive statistics

In this section, we embark on a systematic exploration of the dataset to clean insights into its underlying structures and patterns, focusing on three pivotal attributes: age, delivery time, and price. Descriptive statistics serve as a powerful tool in our analysis, enabling us to summarize and comprehend the data's essential characteristics and tendencies. This foundational analysis is instrumental in guiding subsequent inferential statistical endeavours and data modelling.

Our examination is meticulously comprehensive, incorporating a wide array of statistical measures to furnish a robust and detailed perspective of the data's distribution. We start by determining the total number of observations (n_{obs}) to understand the dataset's breadth and scope. To assess the diversity and uniqueness within the data, we calculate the number of distinct values ($n_{distinct}$) present in each feature. Recognizing the potential influence of missing or incomplete data on our analysis, we also ascertain the number of missing values ($n_{missing}$), ensuring our conclusions are drawn from a sound and reliable data foundation.

A central aspect of our analysis involves delving into the data distribution of the selected attributes. Through this, we aim to uncover the common tendencies and variability inherent in the data, providing a nuanced understanding of its general behaviour. Key statistical measures such as the mean and standard deviation (SD) offer insights into the data's central tendency and dispersion, helping us gauge its overall pattern and spread.

Additionally, we scrutinize the data's extremities by identifying the minimum and maximum values, which aids in understanding the range and the potential outliers or anomalies present in the data. In summary, this section is devoted to unearthing meaningful patterns and characteristics of the dataset, laying a solid groundwork for further analysis and interpretation.

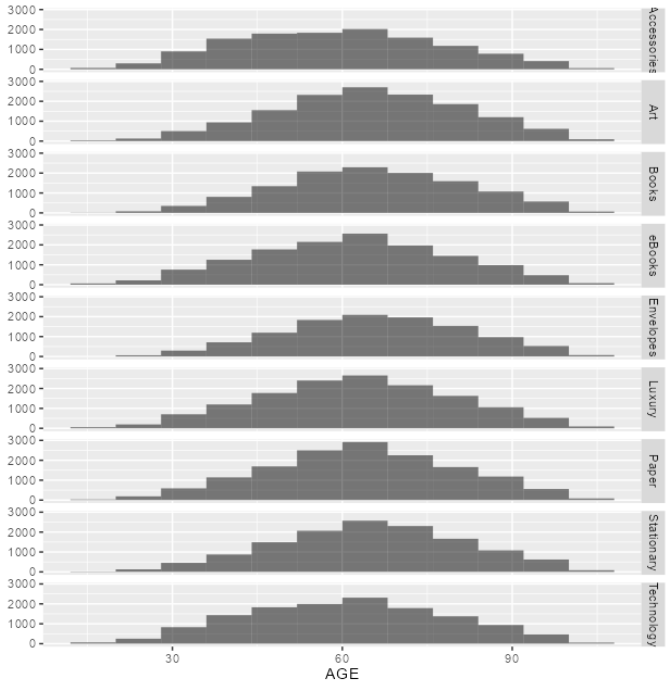
Descriptive statistics of Age

With the information provided in the table below we can see that there are ages of 104 in all departments, Values much higher than that would have been alarming, since the age of 104 is not unheard of we will consider it as a valid age.

DEPARTMENT	variable	n_obs	n_distinct	n_missing	sd	mean	min	max
All	All	All	All	All	All	All	All	All
Accessories	AGE	12,480	89	0	18.051	59.787	16	104
Art	AGE	14,296	89	0	16.532	65.317	16	104
Books	AGE	12,279	88	0	16.199	65.794	17	104
eBooks	AGE	13,739	89	0	17.319	62.283	16	104
Envelopes	AGE	11,267	85	0	15.981	66.305	20	104
Luxury	AGE	14,440	89	0	17.048	63.091	16	104
Paper	AGE	14,796	89	0	16.644	63.971	16	104
Stationary	AGE	13,356	89	0	16.437	65.436	16	104
Technology	AGE	13,347	89	0	17.758	61.223	16	104

Distribution of age over the departments

Here we have the distributions of age over the 9 different departments, the distribution of Accessories is skewed to the right and therefore we can assume that accessories and technology are more appealing to the younger age groups. The rest of the apartments have a triangular distribution, and the greatest appeal is to the age group of 60 to 70 years of age.



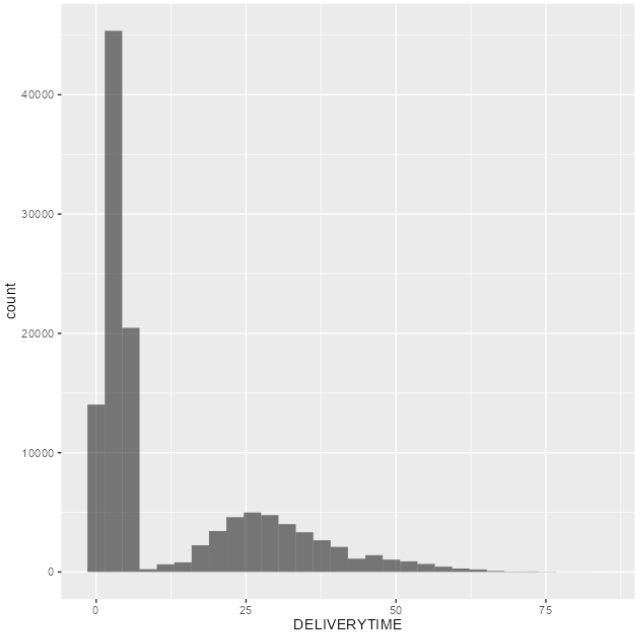
Descriptive statistics of Delivery time

We can see in some departments there are very large delivery times, and this might be a problem regarding customer satisfaction since these deliveries might be late.

DEPARTMENT	variable	n_obs	n_distinct	n_missing	sd	mean	min	max
All	All	All	All	All	All	All	All	All
Accessories	DELIVERYTIME	12,480	59	0	7.941	30.016	1	63
Art	DELIVERYTIME	14,296	8	0	1.045	4.004	1	8
Books	DELIVERYTIME	12,279	8	0	1.037	4.008	1	8
eBooks	DELIVERYTIME	13,739	2	0	0.083	1.007	1	2
Envelopes	DELIVERYTIME	11,267	8	0	1.035	4.006	1	8
Luxury	DELIVERYTIME	14,440	45	0	5.972	23.956	1	48
Paper	DELIVERYTIME	14,796	8	0	1.042	3.992	1	8
Stationary	DELIVERYTIME	13,356	9	0	1.034	4.004	1	9
Technology	DELIVERYTIME	13,347	82	0	12.062	40.134	1	85

Distribution of delivery times

Here we can see the higher delivery times form a skewed normal distribution to the right, with inspection we might be able to reduce the spread into a more uniformly shaped normal distribution.



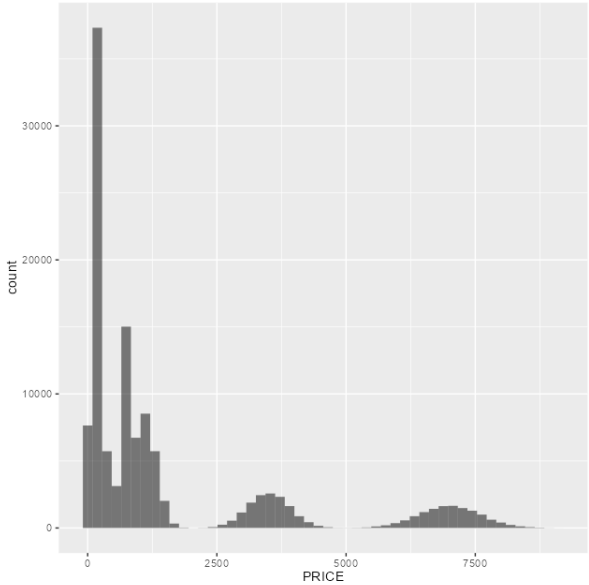
Descriptive statistics of price

Analysing the table allows us to see high value department and the average purchasing price of the department, this can be used in determining the pricing on promotions and the margins available to similar products.

DEPARTMENT	variable	n_obs	n_distinct	n_missing	sd	mean	min	max
All	All	All	All	All	All	All	All	All
Accessories	PRICE	12,480	1,074	0	198.752	1,200.779	375	2,018
Art	PRICE	14,296	1,095	0	199.873	998.871	194	1,821
Books	PRICE	12,279	126	0	17.979	249.847	173	315
eBooks	PRICE	13,739	176	0	25.112	108.032	9	207
Envelopes	PRICE	11,267	106	0	15.075	99.967	37	157
Luxury	PRICE	14,440	2,025	0	402.477	3,502.375	2,056	4,977
Paper	PRICE	14,796	326	0	50.156	699.747	496	890
Stationary	PRICE	13,356	204	0	30.126	270.201	151	398
Technology	PRICE	13,347	2,755	0	598.436	6,999.056	4,663	9,124

Distribution of price

It is clear that there are gaps within the pricing of products thus new departments, or a wider price range can be adopted to cover a bigger range of the market to increase sales.



Section 3

Control charts for various departments

In this section, we present meticulously calibrated control charts for various departments, constructed to facilitate a nuanced analysis of process stability and performance over time. Our methodological approach involved an initial calibration phase, utilizing the first 30 samples. Each sample consisted of 19 instances, establishing a solid baseline for subsequent analysis.

The control charts are accurately designed for clarity and ease of interpretation. Distinctive red lines delineate the 2-sigma control limits, serving as critical boundaries in identifying significant deviations or outliers in the process. Complementarily, the green lines demarcate the 1-sigma control limits, providing additional granularity in evaluating the process variability and consistency.

A central black line represents the process mean or centre line, serving as a reference point for assessing process variations. Accompanying this are the Upper Control Limit (UCL) and Lower Control Limit (LCL), also represented by black lines, which together define the acceptable range of process variation.

Visual markers enhance the charts' interpretability, with red dots pinpointing the samples that transgressed the UCL and LCL, signalling potential outliers or instances of significant process variation. Additionally, yellow markers illuminate instances of violating runs, highlighting sequential data points that display unusual or non-random patterns of variation.

Through these comprehensive control charts, we aim to offer a vivid and insightful visual representation of process behaviours across departments, enabling the identification of potential areas for improvement or further investigation.

eBooks

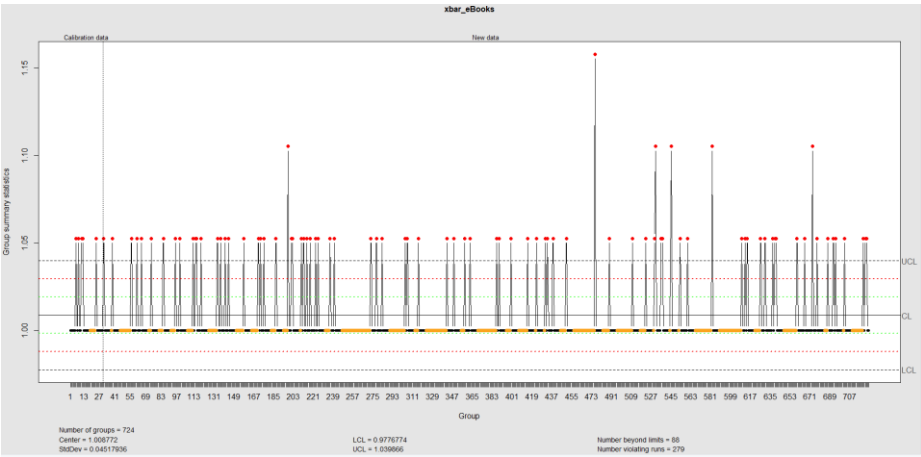
The current dataset presents challenges in capturing a rich and nuanced understanding of the underlying phenomena due to its limited variability and infrequency of data changes. A lack of sufficient variability hinders the formation of a well-defined normal distribution, which is crucial for various statistical analyses and interpretations.

This issue seems to stem from the precision of the measurements taken. Our data has been aggregated at a relatively coarse level, with measurements recorded in hours. This level of aggregation appears to obfuscate the finer nuances and variations in the data, leading to a somewhat vague and imprecise reflection of the underlying trends and patterns. (Schalkwyk, 2023)

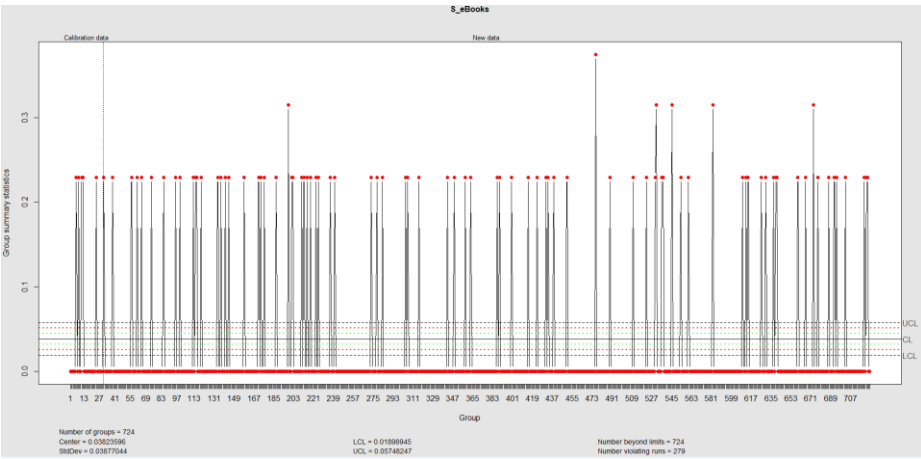
To enhance the richness and accuracy of our dataset, a modification in the data collection approach is proposed. Consider adopting a more finely tuned measurement frequency, such as capturing data in seconds, as opposed to hours. Such an adjustment would likely unveil a more detailed and variable dataset, allowing for a more robust and insightful statistical analysis. This improvement aims to foster a clearer, more precise depiction of trends, patterns, and variabilities, facilitating more informed decision-making and analysis.

Xbar-chart

Samples beyond limits: 6, 8, 11, 719, 721, 722



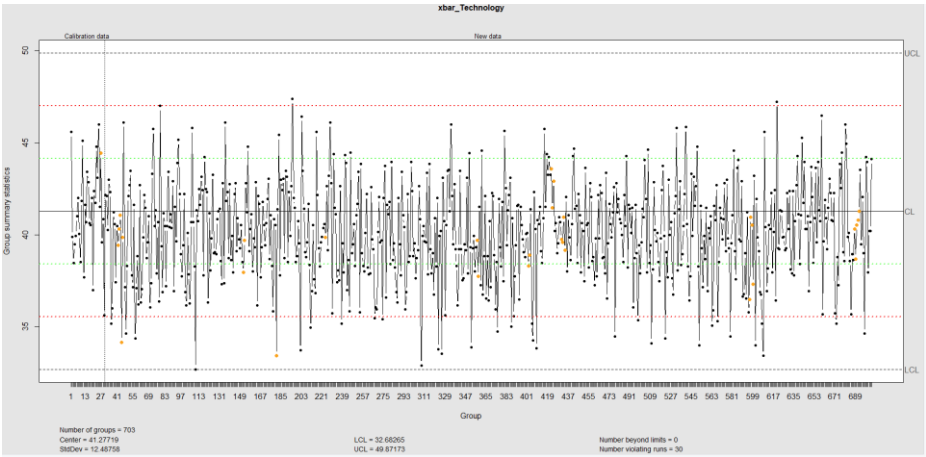
S-chart



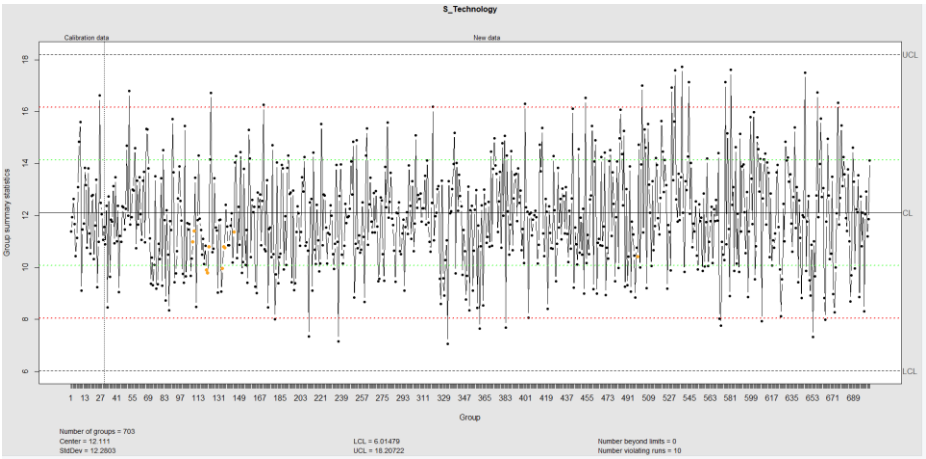
Technology

While evaluating the control charts, it was noted that all samples remained within the acceptable range, as none surpassed the established Upper or Lower Control Limits (UCL and LCL). However, it's important to observe that certain samples approached these limits quite closely. Despite this proximity, there isn't a cause for immediate concern or alarm. The data suggests a stable process, but the closeness to the control limits warrants a degree of vigilance to pre-emptively identify and address potential issues that may affect process stability in the future.

Xbar-chart



S-chart



The graphs of the remaining departments can be found in the appendix.

Capability measures

In evaluating the operational capability of various departments, specific capability measures are utilized to ascertain whether the output of their processes aligns consistently within predefined specification limits. These measures are fundamental in assessing the precision and reliability of departmental processes, ensuring that they meet requisite performance standards and quality criteria.

We employ a comprehensive set of capability indices, each offering unique insights into process performance and conformity. The analysis incorporates the mean and standard deviation, providing a foundational understanding of process central tendency and variation. Additionally, key capability indices, including C_p, C_{pu} , C_{pi} and C_{pk} , are utilized to offer a more nuanced perspective.

- C_p (Process Capability Index) gives a broad overview of process capability by evaluating the width of the process variability in relation to the width of the specification limits.
- C_{pu} and C_{pi} focus on the upper and lower specification limits, respectively, offering insights into whether the process mean is closer to one specification limit compared to the other.
- C_{pk} (Process Capability Ratio) provides a robust measure, assessing the minimum distance of the process mean to either specification limit, adjusted for process variability.

Through these capability measures, we aim to cultivate a thorough and insightful evaluation, ensuring that departmental processes are optimized, reliable, and capable of producing outcomes within the specified limits.

Show 10 entries Search:

capability_indices

	DEPARTMENT	mean	sd	Cp	Cpu	Cpl	Cpk
1	Accessories	29.98245614035088	7.975999208676464	0.1671681877654783	-0.9186918389041416	1.253028214435098	-0.9186918389041416
2	Art	4.064912280701755	1.062707228309019	1.254657254430211	1.234296588459194	1.275017920401228	1.234296588459194
3	Books	4.066666666666666	1.014366630081359	1.314449128937129	1.292541643454844	1.336356614419414	1.292541643454844
4	Envelopes	4.010526315789473	1.013040493073593	1.316169829784358	1.312706224969136	1.31963343459958	1.312706224969136
5	Luxury	24.06315789473684	5.874152717692345	0.2269830897173426	-0.9115163023912233	1.365482481825908	-0.9115163023912233
6	Paper	3.977192982456141	1.074307320687535	1.241109790148342	1.248186293337784	1.234033286958899	1.234033286958899
7	Stationary	4.03859649122807	0.9930787522004614	1.342625980446099	1.329670817476883	1.355581143415316	1.329670817476883
8	Technology	41.27719298245614	12.16409504548209	0.1096122094038185	-0.9118966613910655	1.131121080198702	-0.9118966613910655
9	eBooks	1.008771929824561	0.09332879639769559	14.28640874839628	24.969885465947	3.602932030845553	3.602932030845553

Showing 1 to 9 of 9 entries Previous 1 Next

Capable Departments

Each department listed has successfully exhibited a C_p value greater than 1.33, confirming their processes' ability to operate effectively within the defined Lower and Upper specification limits. This accomplishment indicates a robust process capability, signifying a high level of quality and efficiency in their operational outputs.

Capable_departments

	Departments
1	Art
2	Books
3	Envelopes
4	Paper
5	Stationary
6	eBooks

Process out of and under control

In the subsequent table, a particular presentation of process control across various departments is provided. It specifically highlights instances where samples have deviated from established control limits, signifying areas where processes might be out of control. Each entry facilitates a nuanced appreciation of the process performance, allowing for the identification of occurrences that exceed expected thresholds and may necessitate further investigation or corrective action.

Additionally, the table delineates sequences of consecutive samples falling below the 1-sigma line. This data is enriched with specific details marking the initiation and termination of these sequences, offering insight into the persistence and extent of deviations. This structured presentation aims to foster a clear and comprehensive understanding of process consistency and control status across the evaluated departments.

	Out_Of_Control_Samples	Consecutive_Control_Samples	Start_Sample	End_Sample
Luxury	115, 351, 470, 588	30	35	64
Paper	215, 387	24	33	56
Art	135, 573, 583, 721	24	463	486
eBooks	6, 8, 11, 719, 721, 722	31	241	271
Books	59, 186, 10, 559, 587, 645	37	144	180
Envelopes	461, 161, 474	34	219	252
Accessories	317	40	351	390
Stationary	177, 579	29	659	687
Technology		34	443	476

Section 4

Type I Error in Control Charts

In the realm of control charts, a Type I error, commonly referred to as a false positive or alpha error, denotes the probability of mistakenly inferring that a process is out of control when it remains in control. This error is intrinsically linked to the control limits delineated on the chart.

The rate of Type I error in control charts is conventionally determined by these control limits. The most prevalent control limits are demarcated at ±3 standard deviations from the centreline. Given data that follows a normal distribution, these limits correspond to a Type I error rate of approximately 0.0027 for each side (both upper and lower). Cumulatively, this results in a total error rate of around 0.0054.

Considering the specific scenarios outlined:

A. Single Xbar Sample Beyond Outer Control Limits:

When a single data point is observed outside the 3-sigma control limits purely by chance (with the underlying process being in control), the likelihood stands at roughly 0.27% (or 0.0027) for each side. This percentage represents the Type I error rate for this particular rule.

B. Consecutive Xbar Samples Between-1 and +1 Sigma-Control Limits:

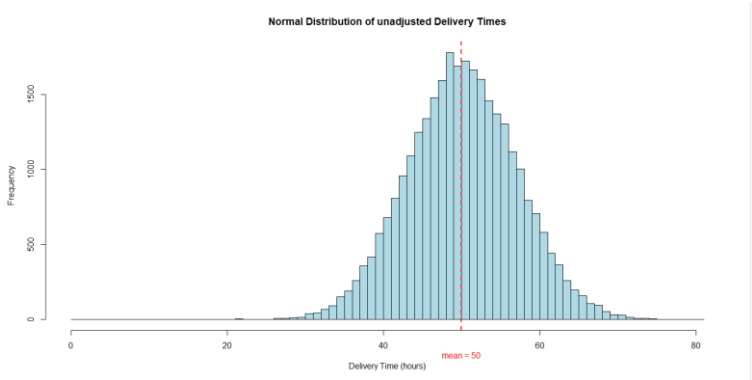
This rule introduces a higher degree of complexity. For a single data point, the probability of it residing between -1 and +1 sigma is approximately 68.27%, a figure derived from the properties inherent to the normal distribution. For a sequence of **k** consecutive points all falling within this range, the probability is calculated as (0.6827) ^k. To ascertain the Type I error for this rule, additional details are necessary, such as the exact count of consecutive points under consideration.

Optimising delivery times

In a targeted effort to enhance delivery efficiencies and reduce associated expenses, a methodical revision of our delivery process was undertaken. Faced with a hefty penalty of R500 per product for each hour of delay, a thorough evaluation and systematic adjustment of the mean delivery time were executed with precision. The objective was to diligently reduce the instances of tardiness and the consequential penalties. A strategic approach was adopted, where the mean delivery time was incrementally decreased by one hour, continuously fine-tuning the process until an optimum balance was achieved, all at a manageable operational cost of R1.59 per product per hour. This meticulously crafted strategy aimed at minimizing the overall expenditure, focusing on diminishing the prevalence of late deliveries and their significant financial repercussions, thereby realizing a refined, optimized, and cost-efficient delivery process.

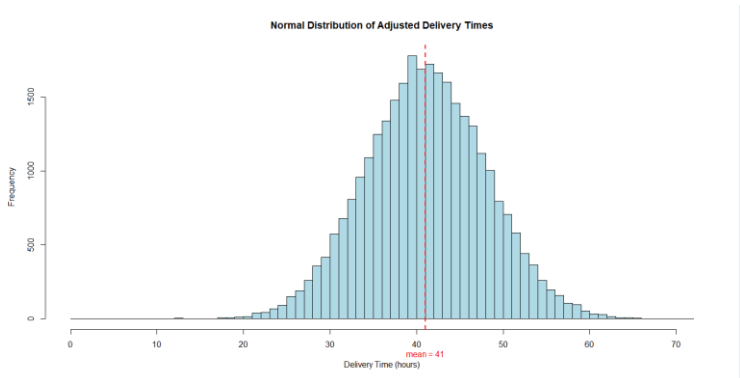
Normal distribution of unadjusted mean

In the accompanying plot, the initial distribution of delivery times is vividly illustrated, allowing for a comprehensive visualization of the original patterns and tendencies in the data.



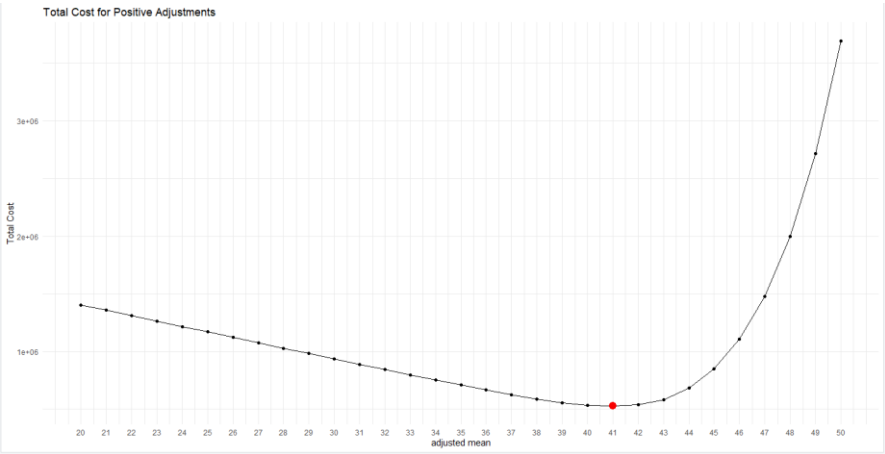
Normal distribution of adjusted mean

n the subsequent plot, the distribution of delivery times, adjusted for optimization, is prominently displayed. This visualization illuminates the enhancement achieved through strategic adjustments, showcasing the new, optimized mean as a pivotal element in our refined delivery process. The adjustments made manifest as a more favourable distribution, underscoring our meticulous efforts to improve delivery times and overall operational efficiency.



Visualization of minimum cost

Below is a visualization that succinctly illustrates the optimization of our delivery process to minimize costs. The line plot displays the relationship between adjusted mean delivery times and their associated costs, with a distinctive red dot highlighting the point of minimum cost. The x-axis clearly indicates the optimal mean, showing the most cost-efficient adjustment identified in our analysis.



Optimal reduction in hours and total minimum cost

Below is a table summarizing the results of our optimization process. It includes the "Optimal i" indicating the number of iterations required to find the most effective mean. The table also displays the minimum total cost achieved, as well as the optimal mean itself, providing a concise overview of the efficiency and financial benefits realized through our optimization efforts.

	Parameter	Value
1	Optimal i	9
2	Minimum Total Cost	526587.04966
3	Optimal mean	41.09038885120967

Section 5

Single factor experiment

In this section, we present the findings from the hypothesis tests conducted to investigate the differences in delivery times across various departments and over different years.

Experiment 1: Comparing Delivery Times Across 9 Different Departments

All departments

Single Factor Experiment Results for departments

	V1	V2	V3	V4	V5	V6
1	Source	SS	DoF	MS	fo	P-value
2	Treatment	223.68	8	27.96	1.01	0.4252831
3	Error	2805448.18	101394	27.67	---	---
4	Total	2805671.87	101402	---	---	---

-The calculated p-values in this experiment were all larger than the chosen significance level (α = 0.025).

We fail to reject the null hypothesis. This suggests that there are no statistically significant differences in the mean delivery times across the nine departments.

Experiment 2: Comparing Delivery Times Across 5 Different Years for Each Department

Luxury

Single Factor Experiment Results for Luxury

	V1	V2	V3	V4	V5	V6
1	Source	SS	DoF	MS	fo	P-value
2	Treatment	152.15	4	38.04	1.39	0.2343293
3	Error	266796.54	9755	27.35	---	---
4	Total	266948.69	9759	---	---	---

Similarly, the p-values obtained in this experiment were also larger than the significance level (α = 0.025).

We fail to reject the null hypothesis. This indicates that there are no statistically significant differences in the mean delivery times across the five years for each department.

None of the tests conducted rejected the null hypothesis. The p-values calculated were consistently larger than the chosen significance level, indicating that the observed differences in delivery times among the departments and across the years are likely due to random variation rather than a systematic effect.

Least significant difference

The Least Significant Difference (LSD) is a method used in conjunction with Analysis of Variance (ANOVA) to conduct pairwise comparisons between group means to determine if they are significantly different from each other. LSD is used when you have a significant overall ANOVA result, and you want to identify which group(s) is/are different from the others.

All departments

LSD Results for departments

	Treatment 1	Treatment 2	Treatment 3	Treatment 4	Treatment 5	Treatment 6	Treatment 7	Treatment 8	Treatment 9
1	0	0.036123191621550 53	0.1328658915416714	0.1586047750066566	0.07198011893139267	0.03221798171651713	0.07251264755480591	0.09736398331410356	0.08129936984112973
2	0	0	0.0967426999201208 6	0.1224815833851061	0.03585692730984213	0.0039052099050334 02	0.03638945593325538	0.06124079169255303	0.0451761782195792
3	0	0	0	0.02573888346498521	0.06088577261027872	0.1006479098251543	0.06035324398686548	0.03550190822756782	0.05156652170054166
4	0	0	0	0	0.08662465607526393	0.1263867932901395	0.08609212745185069	0.06124079169255303	0.07730540516552686
5	0	0	0	0	0	0.03976213721487554	0.0005325286234132 42	0.0253838643827109	0.0093192509097370 65
6	0	0	0	0	0	0	0.04029466583828878	0.06514600159758643	0.0490813881246126
7	0	0	0	0	0	0	0	0.02485133575929765	0.0087867222863238 23
8	0	0	0	0	0	0	0	0	0.01606461347297383
9	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0
13	0	0	0	0	0	0	0	0	0
14	0	0	0	0	0	0	0	0	0
15	0	0	0	0	0	0	0	0	0
16	0	0	0	0	0	0	0	0	0
17	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0

18 records

Paper

LSD Results for Paper

	Treatment 1	Treatment 2	Treatment 3	Treatment 4	Treatment 5
1	0	0.1330624682580002	0.3164042661249358	0.009141696292534363	0.09090909090909172
2	0	0	0.4494667343829359	0.1239207719654658	0.04215337734890845
3	0	0	0	0.3255459624174701	0.4073133570340275
4	0	0	0	0	0.08176739461655735
5	0	0	0	0	0
6	0	0	0	0	0
7	0	0	0	0	0
8	0	0	0	0	0
9	0	0	0	0	0
10	0	0	0	0	0

10 records

After looking at the results it is evident that there are no large differences within the data further justifying our results from the single factor experiment.

Section 6

Reliability of service

In the subsequent section of this report, we delve into a strategic analysis aimed at minimizing annual operational costs related to employee hiring and penalties incurred due to insufficient staffing. A critical decision revolves around determining the optimal number of employees to hire, which directly influences our overall costs.

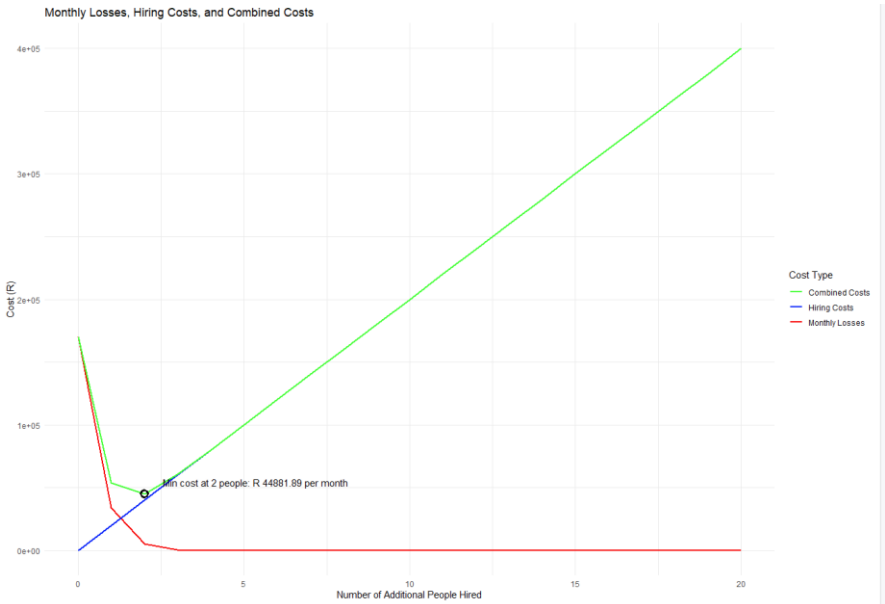
Given the parameters, a penalty of R17 000 is imposed for each day less than 14 employees are present. Conversely, the option to hire additional employees is available at a cost of R20 000 per employee per month. To navigate this decision-making process, we have employed the Binomial Theorem as a fundamental analytical tool.

The Binomial Theorem is a powerful mathematical principle used to expand expressions that are raised to a power. In our context, it is applied to model different scenarios and probabilities related to employee attendance. By considering the probabilities of various attendance levels and their associated costs or penalties, we can compute and compare the total expected costs under different hiring strategies. This approach allows us to systematically evaluate the trade-offs and identify the hiring strategy that minimizes the total expected annual cost.

Through this meticulous application of the Binomial Theorem, we aim to unveil a data-driven, cost-efficient strategy for employee hiring, thus enhancing operational effectiveness and financial sustainability.

Optimal hiring solution for minimum cost

Here we can see that an optimal minimum lies at an additional 2 employees to be hired, thus bringing our minimum cost per month to just R44 881.89.



P-value

In our strategic approach to minimizing annual operational costs, an iterative process was diligently employed to identify the most advantageous probability value, denoted as p. Through a series of systematic iterations, we meticulously explored various p values to discern their impact on employee attendance and associated costs.

The iterative exploration culminated in the identification of an optimal p value of 0.9717. This value was instrumental in guiding our hiring strategy, aligning with our objective of cost minimization. Utilizing this optimal p value, we were able to forecast attendance, resulting in predicted days of 0, 0, 0, 4, 37, 198, and 490. These predictions serve as a valuable output, facilitating informed decision-making aimed at optimizing our hiring approach to achieve enhanced operational efficiency and cost-effectiveness.

Category	Value
Estimated Value of p	0.9719
Predicted Days (for each employee count)	0, 0, 0, 4, 37, 198, 490

Results of optimal Solution

Category	Value
Minimum Yearly Cost (R)	538582.62
Expected Days per Year More Than 14 People	338

Conclusion

In conclusion, the meticulous analysis conducted on the sales data using Statistical Process Control (SPC) has unveiled several pivotal insights that are instrumental for enhancing operational efficiency and customer satisfaction. The initial stages of data wrangling ensured a robust foundation, enabling a comprehensive exploration of various attributes such as age, delivery time, and price in the subsequent sections.

Descriptive statistics revealed nuanced patterns and tendencies, such as the appeal of certain departments to specific age groups and the identification of potential issues related to delivery times. Control charts further facilitated a nuanced understanding of process stability across various departments, enabling the identification of significant deviations and outliers that warrant further investigation or improvement.

The study's findings, such as the identification of gaps within the pricing of products and the potential to optimize delivery times, serve as valuable inputs for strategic decision-making. These insights pave the way for informed strategies aimed at optimizing pricing, improving delivery times, and ultimately enhancing customer satisfaction and operational excellence.

References

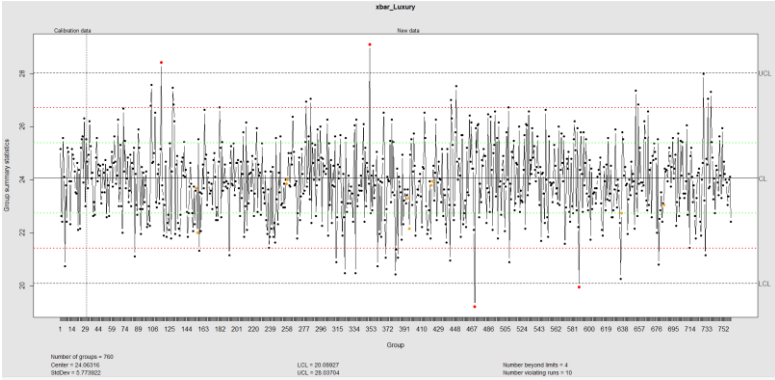
Schalkwyk, T. G. (2023, 09). *Sunlearn.* Retrieved from Quality Assurance.

Appendix

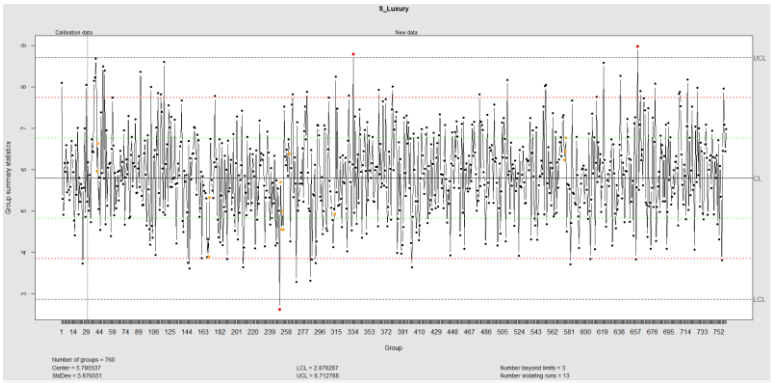
Luxury

Xbar-chart

Samples Beyond limits: 115, 351,470, 588



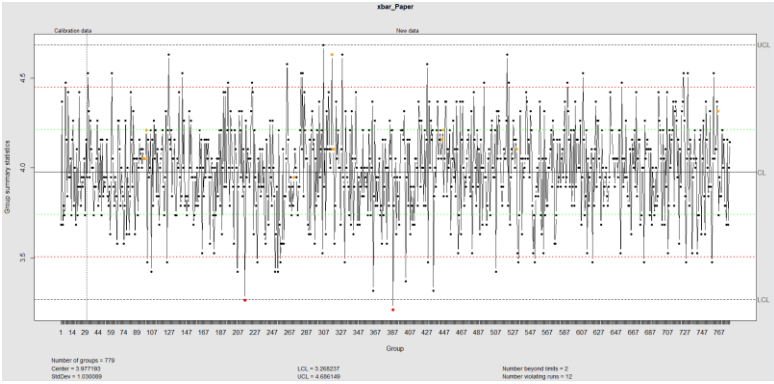
S-chart



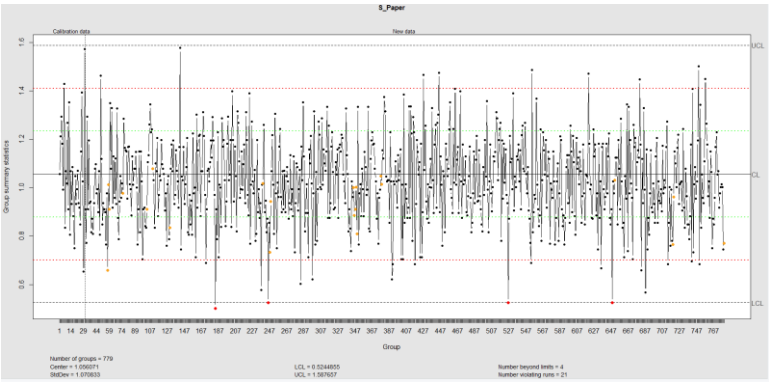
Paper

Xbar-chart

Samples Beyond limits: 215, 387



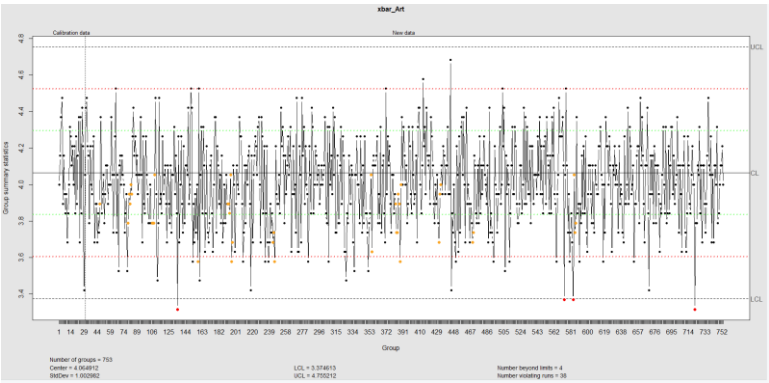
S-chart



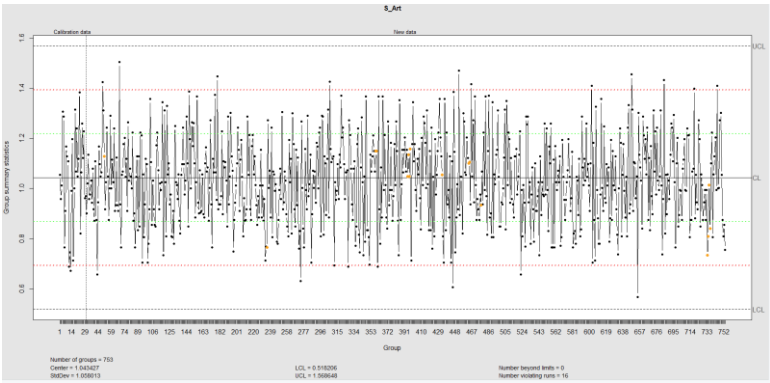
Art

Xbar-chart

Samples Beyond limits: 135, 573, 583, 721



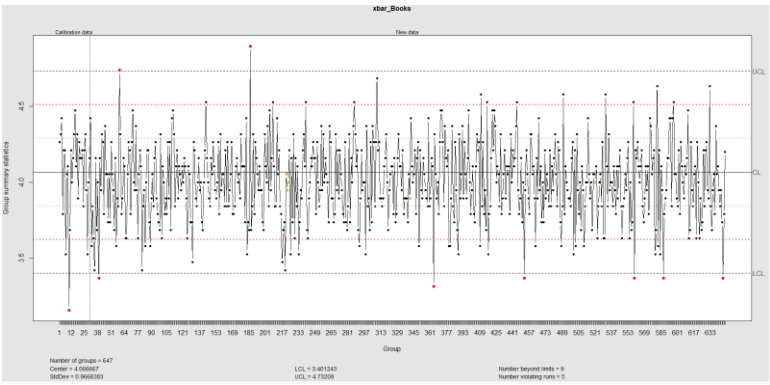
S-chart



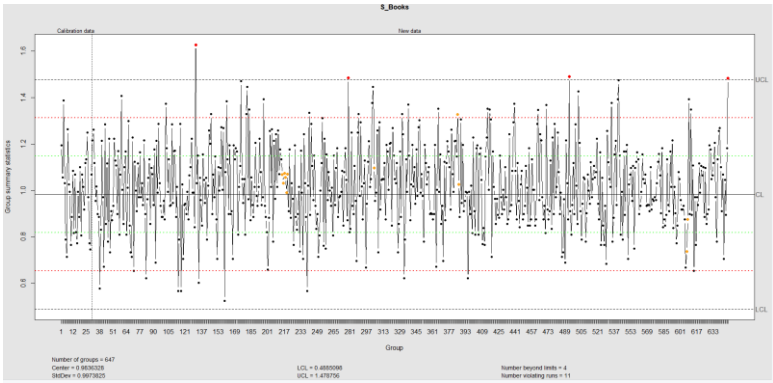
Books

Xbar-chart

Samples Beyond limits: 10, 59, 186, 559, 587, 645



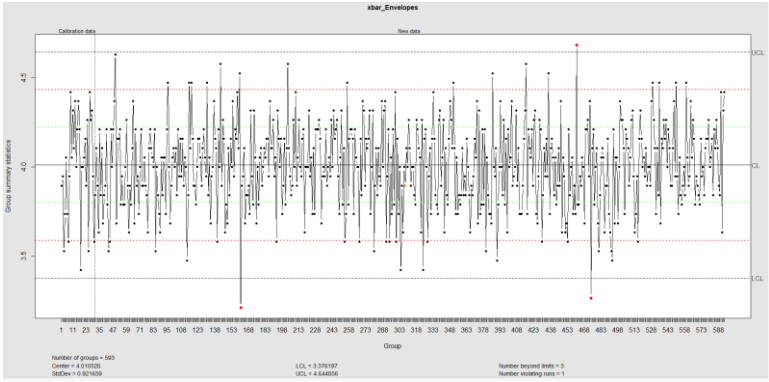
S-chart



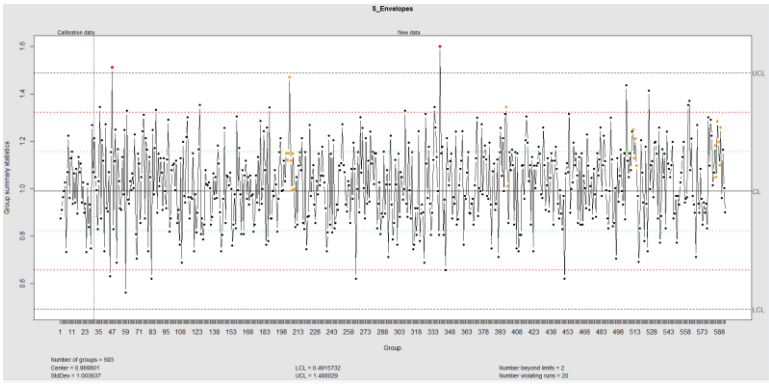
Envelopes

Xbar-chart

Samples Beyond limits: 161,461,474



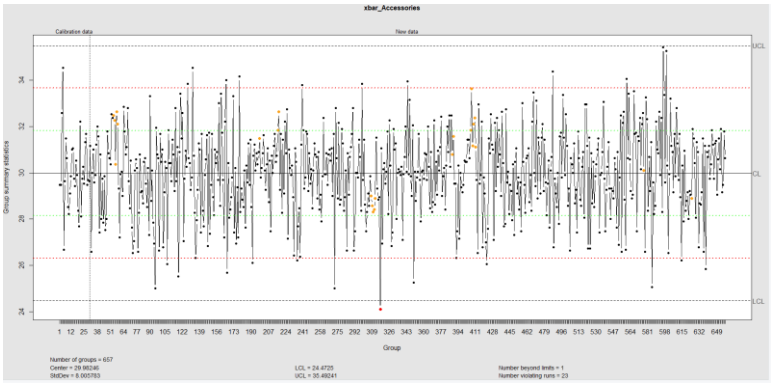
S-chart



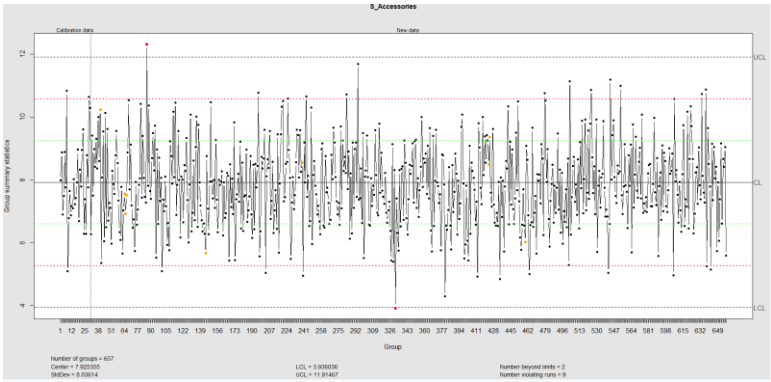
Accessories

Xbar-chart

Samples Beyond limits: 317



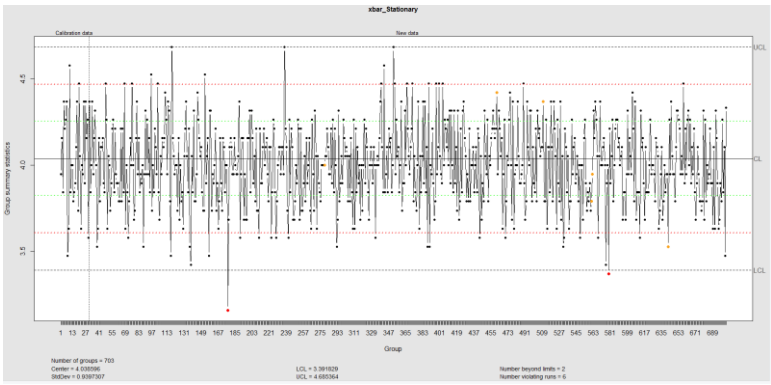
S-chart



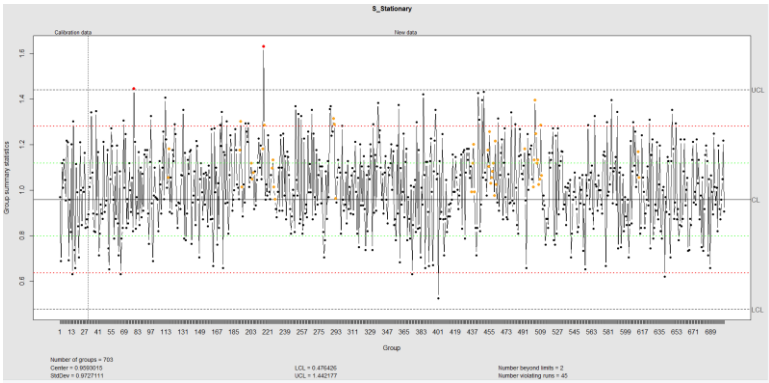
Stationary

Xbar-chart

Samples Beyond limits: 177, 579



S-chart


S_Stationary
Calibration data
New data
Group summary statistics
UCL
CL
LCL
Group
Number of groups = 703
Center = 0.9593015
StdDev = 0.9727111
LCL = 0.476426
UCL = 1.442177
Number beyond limits = 2
Number violating runs = 45